SIGMA fp
Dual Base ISO Technology

Takuma Wakamatsu

SIGMA Corporation

October 10th, 2020

Introduction

SIGMA fp offers so-called Dual Base ISO technology, which allows you to use ISO settings with two base sensitivities. A general digital camera that has only one base sensitivity has a problem that noise increases when it is set to high sensitivity. There was also the issue of narrowing the dynamic range. Dual Base ISO technology has two base sensitivities, and when set to high sensitivity, the high sensitivity of the base sensitivity can be used to keep noise low and wide dynamic range.

This technology is realized by using the Dual Base ISO circuit mounted on the image sensor. SIGMA fp did not initially use this feature. In order to minimize noise in all ISO settings, we took time to tune and activated with FW ver.2.0.

With this technology, it is possible to shoot images with less noise even when shooting in low light. Also, by making the best use of the dynamic range of the camera, you can increase the flexibility in post-production and improve the quality of your project.
Furthermore, by improving the quality of the footage, the man-hours of post-production such as noise reduction will be reduced, which will contribute to the cost reduction of the project.

Details

The optimum base sensitivity is set for each shooting mode. The ISO setting is optimized to ensure image quality. Base ISO of each shooting mode is as followings:

Still Mode

- JPEG, DNG 100/640

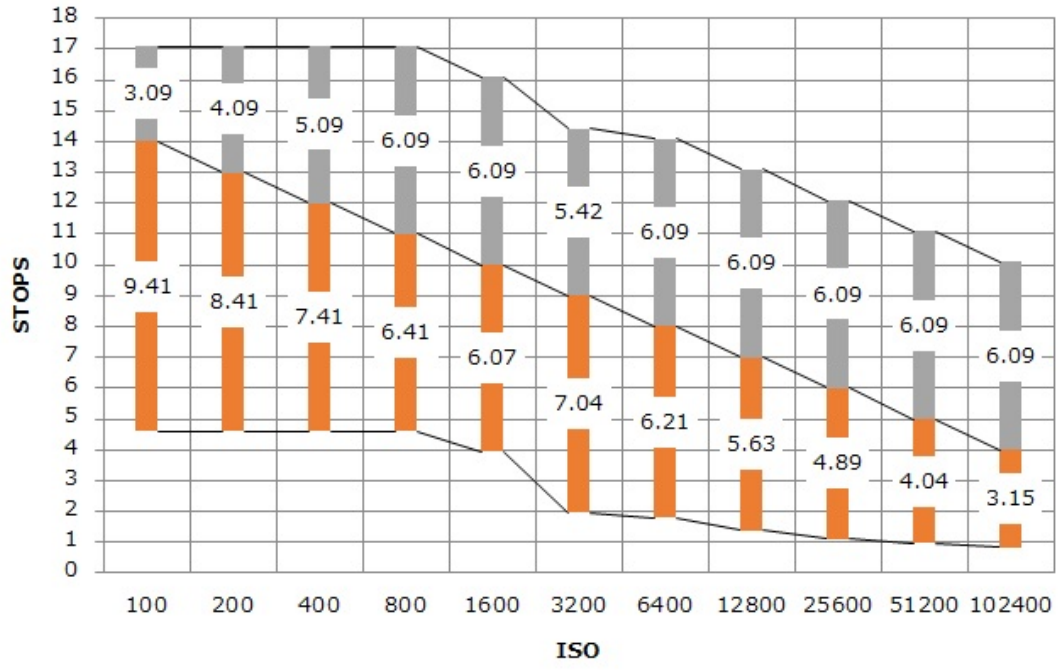
CINE Mode

- Cinema DNG 12bit, HDMI RAW 100/3200
- MOV, Cinema DNG 10, 8bit 100/640

The base ISO switches automatically when you choose ISO. There is no need for the operator to switch manually. As usual, the optimum base sensitivity is selected by simply changing the ISO, and the best image quality is provided.

Tips

Which ISO should I choose for the actual shooting? Here, we will explain practical ISO setting tips using a dynamic range chart. This knowledge will be very useful especially for serious users who want to maximize the dynamic range of the camera in RAW recording.



This dynamic range chart is when Cinema DNG 12bit and HDMI RAW are set. Base ISO is 100 and 3200.
The maximum dynamic range of SIGMA fp is 12.5 stops. You can see that there are 12.5 stops at ISO 100 to 800 and 3200.

The boundary between the gray bar and the orange bar is the exposure where 18% gray is depicted as 18% gray, which is the so-called standard exposure. The gray bar is the dynamic range from standard exposure to clipped white, the so-called headroom. The orange bar is the dynamic range from standard exposure to crushed shadows, the so-called leg room.

From ISO 100 to 800, you can get the maximum dynamic range of 12.5 stops no matter which ISO you choose, but the ratio of headroom to leg room is different.

If shadow detail is more important than highlight roll-off, ISO 100 is a good choice. Even shadow details that don't appear on the display are well recorded in the 9.41stops legroom. You can recover the details by applying gain to the dark areas with grading software in post-production. However, ISO 100 headroom has only 3.09 stops. Please note that areas brighter than 3.09 stops from the standard exposure will also be clipped.

If you want to use ISO100, you can set the exposure darker and increase the exposure with post-production to suppress clipped white.

If you value highlights or are facing the problem of clipped white easily, it's best to use ISO 800, which has plenty of headroom, even in bright environments. In this case, the exposure will be 3 steps brighter than ISO100, so adjust the exposure appropriately with the Iris of the lens or ND filter. ISO800 has a dynamic range of 6.09 stops in headroom and 6.41 stops in leg room, providing a maximum dynamic range of 12.5 stops, which is the same as ISO 100 with base ISO. The setting is very well-balanced, so it is recommended for all kinds of shooting.

Which ISO would you choose if you wanted to use a higher sensitivity than ISO800? We recommend ISO 3200.
ISO1600 maintains the same headroom as ISO800, but the legroom is reduced from 6.41 to 6.07 and the dynamic range is 12.16stop. ISO 1600 is not the best choice for low light shooting where shadow detail is important.

ISO3200 is the base ISO on the high sensitivity side. The leg room has 7.04 stops, which is enough to record shadow details. It also has a maximum dynamic range of 12.5 stops, which is the same as ISO 100 which is the base ISO on the low sensitivity side. If you are uncertain about choosing an ISO for low light shooting, it is a good idea to choose ISO 3200.

ISO6400 has a headroom of 6.09 stops and a leg room of 6.21 stops, making it a well-balanced setting. The dynamic range is 12.3stops, which is not a bad choice for general purpose.

Headroom 6.09 stops is kept above ISO 12800. However, it should be noted that as the ISO is raised, the leg room will gradually decrease. Don't expect shadows to be recorded as in ISO 3200. It cannot be fixed by post-production. If you want to emphasize the details of the shadow, you need to take some measures such as shooting with a bright exposure in advance to secure the leg room, or shining a light on the shadow in advance to leave no shadow during shooting.

Summary

With the spread of HDR content, the dynamic range and noise level required for footage have become better. In particular, the reproduction of shadow details in low-light shooting is getting attention. This instruction gave a hands-on explanation of the SIGMA fp ISO settings.
Choosing an ISO with a deep understanding will lead to better shooting of footage.
We hope that this Dual Base ISO technology will help SIGMA fp create many beautiful images and contribute to your project to be better.